Human-LLM Co-Pilot

Driver preferences, moving traffic, one continuous task clock.

The driver selects **Overtake** and says, **"I prefer smoothness and energy efficiency."** A slower lead vehicle and a passing car constrain the maneuver. Language and brief evaluations guide planning within one **continuous 33 s task window**.

DRIVER
Sets the overtake goal; expresses preferences and evaluates plans.

LOCAL LLM
Qwen 9B + source binding identifies preferences for the current task.

MODEL + PLANNER
Updates preferences; checks traffic; computes plans and task energy.

CONTROLLER
Executes the selected plan in the native traffic simulation.

01 The overtaking situation

Initial state | ego 72 km/h

Five surrounding vehicles are present in both native simulation and planning. Initially, the four immediate candidates fail the passing-car clearance check; the traffic gap opens as that car moves ahead. Markers enlarged; gaps relative to ego.

02 Same goal, two predicted execution paces

Candidates at 1.9 s

CANDIDATE	FINISH	TRANSITION	TASK ENERGY	PEAK / COST
A Earlier finish	26 s	3.5 s	**69.81 Wh**	94.5 km/h / 32.28
B More time	29 s	3.5 s	**56.51 Wh**	77.1 km/h / 28.20

Cost = planner smoothness cost (lower is better). Energy values are predictions for these plans.

03 Two evaluations lead to a new plan

- **2.5 s** Comparison received: **B is preferred** over A.
- **4.1 s** Acceptance received: a **separate 26 s / 4.5 s candidate** is acceptable.
- **4.7 s** Replan using both evaluations and the current vehicle state: **29 s finish / 4.5 s transition**. Relative to B, finish time stays at 29 s; the transition grows from 3.5 s to 4.5 s.

29 s	56.57 Wh	3 / 4
RECORDED COMPLETION	COMPLETE-TASK MODEL ENERGY	REQUESTS USED

Simulation replay; scripted evaluation feedback; one recorded local Qwen 9B call (~0.989 s).

EXPLORE THE DEMO zv-nus.github.io/human-llm-copilot